

UTBot Java at the SBST2022 Tool Competition

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The motivation behind UTBot Java is to provide an efficient, scalable solution for automatic unit test generation for Java programs. This tool does this by combining symbolic and concrete execution. Its main goal is to increase code coverage, while reducing the need for manual testing. To achieve this, UTBot analyzes the control flow of Java methods and uses SMT solvers to find inputs that satisfy different execution paths, helping it generate effective white-box tests automatically.

What sets UTBot apart is its use of symbolic execution alongside symbolic mocks and concrete execution, offering a more flexible and powerful testing approach. Two main configurations of UTBot were tested: UTBot-concrete, which focuses on minimal use of mocks with concrete execution, and UTBot-mocks, which relies more on symbolic mocks without concrete execution. Features like wrappers for standard library classes, the use of Mockito for symbolic mocks, allowing UTBot to simulate behaviors of external objects and methods, and still generate valid and relevant test cases. On top of that, the team implemented wrappers for complex Java standard library classes like Map, Set and String, to simplify the symbolic analysis, which is a significant achievement given the complexity of the Java ecosystem. They also added a fallback strategy where default test cases are created using concrete execution when symbolic analysis fails or times out, so the tool always tries to generate something, even under tough conditions.

UTBot was evaluated during the SBST 2022 Tool Competition. While both versions faced issues like missing dependencies and compilation errors, UTBot-concrete still ranked third overall, and UTBot-mocks placed sixth. Despite the technical setbacks, the tools achieved line coverage of up to 46.8% and branch coverage of 39.2% within a 120-second time limit. The team's reflections on debugging and improving test generation show a strong focus on refining the tool further. Overall, UTBot Java shows strong potential as a scalable and effective solution for automated testing.